Retail Marketing Analytics on the Census Income Dataset

A classifier for $50k earners and a 6-persona marketing segmentation

Client-facing project report - April 2026.
Dataset: 1994-1995 U.S. Current Population Survey extract, 199,523 records, 40 demographic and employment features, survey-weighted to ~347M people.

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1. Objective & approach

The client asked for two things:

1. A classifier that identifies whether a person earns more or less than $50,000 per year from their demographic and employment profile, so that marketing treatments can be tailored to each income segment.
2. A segmentation of the same population that the client's marketing team can turn into actionable personas for acquisition and retention.

Both are built on the same data pipeline, sit behind the same preprocessing module, and are fit with **population survey weights**, so every number in this report reflects the U.S. population and not just the sampled extract. The two models are intentionally different in intent:

* The classifier is **discriminative**. It tells you *which individuals* are likely to be high-earners, and is evaluated by how sharply it ranks them.
* The segmentation is **descriptive**. It tells you *what kinds of people* exist in the population, regardless of what they earn - which is what marketing teams need when they have to design creatives, channels, and product catalogs, not just target lists.

Trying to fold them into a single model would sacrifice interpretability on the segmentation side and accuracy on the classification side.

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2. Data understanding

The numbers: 199,523 rows, 40 features, split roughly 50/50 across 1994 and 1995. The unweighted positive rate for income >$50k is 6.21%; the *population-weighted* rate is 6.41%. The class is very imbalanced (~15:1).

The survey weights matter. They range from 38 to 18,656 (median 1,618) and represent the number of people in the U.S. population each surveyed person stands for. Their sum, 347,245,892, matches the 1994/95 U.S. population closely. Without the weights every metric is biased toward the sampling frame rather than the real population the client wants to market to.

"Not in universe" is not missing data. 13 of the 40 features contain large fractions of the literal string `Not in universe`, which in CPS jargon means "this question was not applicable to this respondent" (for example, children have no occupation code; people who didn't move have no migration codes). Collapsing this into `NaN` and imputing it would destroy real information - these categories are informative about the respondent's life stage and employment status. We keep them as their own category throughout.

True missingness is concentrated. The migration block is ~50% `NaN` because it is only collected on a sub-sample; country of birth is 1-3% `NaN`. Hispanic origin and previous state are <0.5%. Gradient boosting handles these natively without imputation.

Duplicate rows exist (~30k rows have identical feature profiles). We do not drop them - in a large survey, two households with the same covariates are not a data-quality error, they are evidence of the density of the underlying feature distribution. Dropping them would systematically under-weight the most common profile types.

Key bivariate findings

Figures 1-4 below are generated automatically by `scripts/01_run_eda.py` and live in `reports/figures/`.

* **Age (Figure 1)** - a strong, clearly non-monotonic relationship. P(income >$50k) rises from ~0% in the under-18 group, peaks at ~17% for 45-54-year-olds, and drops back to ~4% after 65. This is where most of the signal lives.
* **Education (Figure 2)** - the single most powerful univariate predictor. Professional-degree holders have a 50%+ positive rate; high-school dropouts sit below 2%. The entire curve is monotonic in years of formal education.
* **Sex and race (Figure 3)** - males are 4x more likely than females to be in the >$50k bucket (10.3% vs 2.6%), and Asian/Pacific Islander and White respondents both sit above 7% while Black and Native American respondents sit below 4%. These raw rates reflect historical 1994/95 income distributions and are discussed in §7 (Fairness).
* **Survey weights** are approximately log-normal (Figure 4). Using them at training time essentially re-weights some populations up by 3-4x versus others, which changes the effective sample size of several subgroups non-trivially.

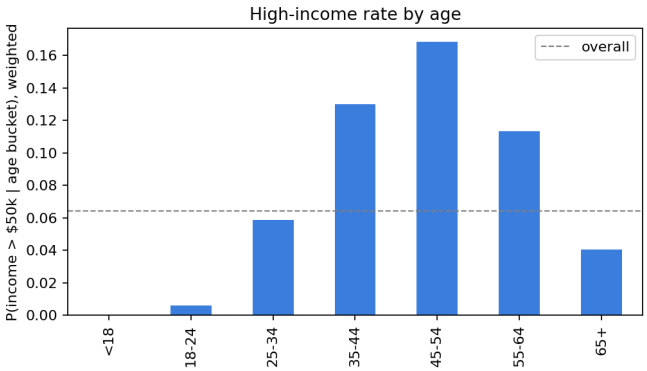

Figure 1 - income rate by age

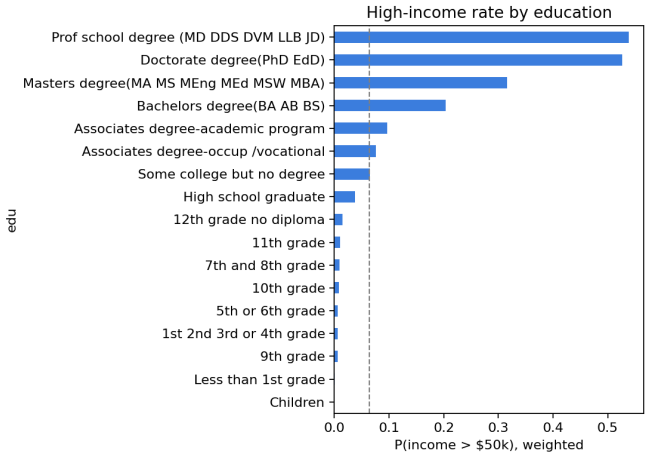

Figure 2 - income rate by education

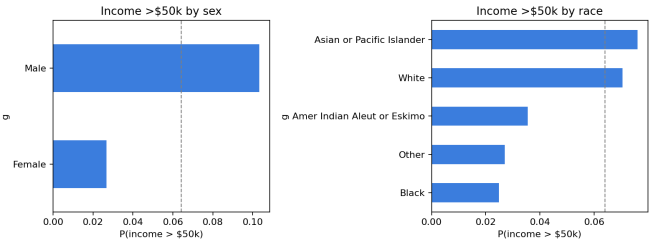

Figure 3 - income rate by sex and race

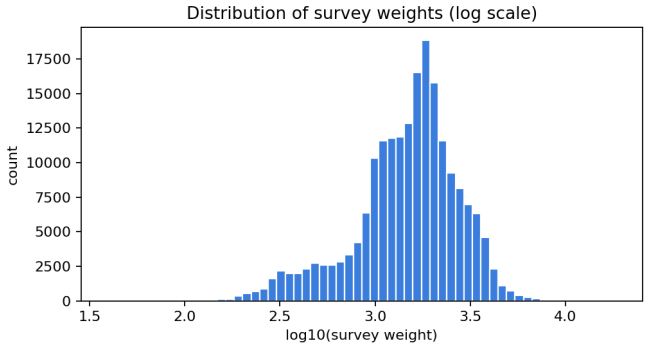

Figure 4 - weight distribution

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3. Preprocessing & feature engineering

Our philosophy: **let the model see the data.** Aggressive feature engineering on a 200k-row tabular problem with a modern gradient-boosting library generally hurts more than it helps; the wins come from *not throwing information away*, not from bolting on derived features.

Concretely:

* **Leading whitespace** is stripped on load. Every cell in the raw file

has a leading space, which silently turns `"Male"` and `" Male"` into two different categories if not handled.

* **`?` becomes `NaN`, `Not in universe` stays as a category.** Different

kinds of missing-ness get different treatment.

* **Numeric vs categorical roles are explicit** in `config.py`.

`detailed industry recode` and `detailed occupation recode` *look* numeric but are BLS code identifiers with no ordinal meaning, so they are categorical. Getting this wrong would corrupt the model with nonsense ordering constraints.

* **Categoricals are cast to `pandas.Categorical`** so that

`HistGradientBoostingClassifier` routes them through its native categorical-split path. No one-hot encoding; no label encoding; no target encoding.

* **Train/test split is 80/20, stratified on the target.** The weights

are preserved in original units across both splits because they are absolute population counts and re-scaling would destroy their meaning. We also union the category sets across splits so that rare countries of birth present on only one side don't become "unseen" at prediction time.

* **For the logistic-regression baseline**, numerics are standardised

and categoricals one-hot encoded with an `"infrequent"` bucket (`min_frequency=20`) so the 47-level occupation code doesn't blow up the design matrix.

No sampling is used to address class imbalance. Instead, the model is fit with the survey weights directly - that's both theoretically cleaner than SMOTE / random oversampling and delivers a natively calibrated probability output, which we verify in §5.

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4. Classification model

4.1 Model selection

Two candidate families:

Logistic regression (baseline). One-hot encoded categoricals + standardised numerics, L2 penalty, `liblinear` solver, survey weights passed via `sample_weight`. This is the "can a linear model solve this?" floor.

Histogram Gradient Boosting (production). `sklearn.ensemble.HistGradientBoostingClassifier` with:

* native categorical-feature support (boolean mask of length 40),
* native missing-value support,
* `log_loss` objective so the output is a calibrated probability,
* early stopping on 15% internal validation,
* L2 regularisation.

HGBC was deliberately preferred over LightGBM and XGBoost here. The three are performance-equivalent on tabular data of this size, and LightGBM / XGBoost require an OpenMP runtime (`libomp`) that adds a brittle system dependency. Since HGBC is bundled with scikit-learn and has the same categorical + missing-value handling, using it keeps the deliverable pip-installable on any machine.

4.2 Hyperparameter tuning

We used **Optuna** with a TPE sampler and a median pruner, tuning on 3-fold cross-validation inside the training split, optimising **population-weighted PR AUC** (not ROC AUC - at 6% base rate PR AUC is the metric that actually reflects the head of the ranking, which is what marketing cares about). 25 trials. Searched space:

parameter	range
`learning_rate`	0.02 - 0.15 (log)
`max_iter`	200 - 900
`max_leaf_nodes`	16 - 96
`min_samples_leaf`	10 - 120
`l2_regularization`	0.001 - 5.0 (log)
`max_features`	0.6 - 1.0

Best configuration: `learning_rate=0.025`, `max_iter=425`, `max_leaf_nodes=77`, `min_samples_leaf=19`, `l2_regularization=1.94`, `max_features=0.68`. Best CV PR-AUC = **0.6669**. The improvement over the hand-picked default (PR-AUC 0.6596) is modest, suggesting that this model family is near-saturated on this data - a useful signal for the client about *how much headroom is left*.

4.3 Threshold selection

Marketing cannot ship a probability, it has to ship a decision. We pick the probability threshold that maximises weighted F1 on **out-of-fol d** cross-validation predictions generated on the training split, so that the test set is completely untouched when the threshold is chosen. The selected threshold is **0.27**.

We report two other operating points too:

* Precision @ top 10% of the ranked list (for ranked mailer targeting);
* Precision @ top 5% (for high-value product recommendations).

4.4 Results

Held-out test set (39,905 rows, weighted to ~69M people):

Metric	LogReg	**HGBC**
ROC AUC	0.9461	**0.9542**

Metric	LogReg	**HGBC**
Average Precision (PR AUC)	0.6264	**0.6931**
Brier score	0.0362	**0.0329**
F1 @ threshold 0.27	0.5915	**0.6225**
Precision @ top 5%	0.6514	**0.7105**
Precision @ top 10%	0.4711	**0.4837**
Lift @ top 10%	7.30x	**7.49x**

What this means in plain English. If marketing takes the top 10% of the ranked population by HGBC score and sends a "high-income" treatment, **48% of them will actually be high-earners** - versus the 6.4% hit-rate of a random mailer. That's a **7.5x lift**. If they tighten to the top 5%, precision rises to 71%. The ROC curve (Figure 5) and PR curve (Figure 6) show both models.

5-fold CV means are within ~0.004 of the test-set numbers with std 0.002 on ROC AUC, so the model has very low variance on this data and is not overfitting the split.

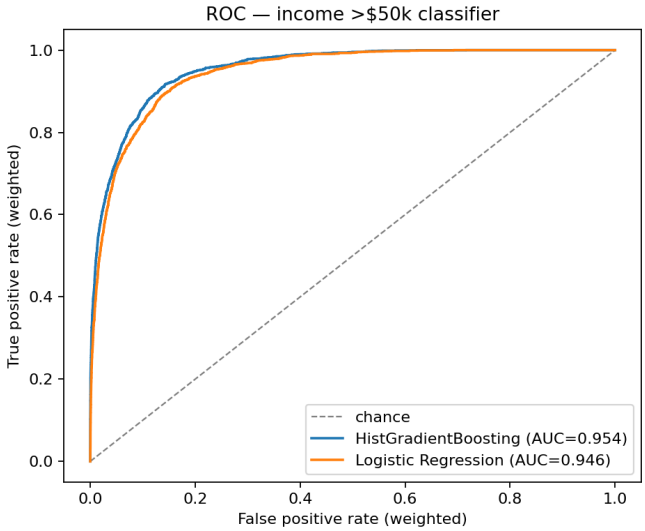

Figure 5 - ROC curves

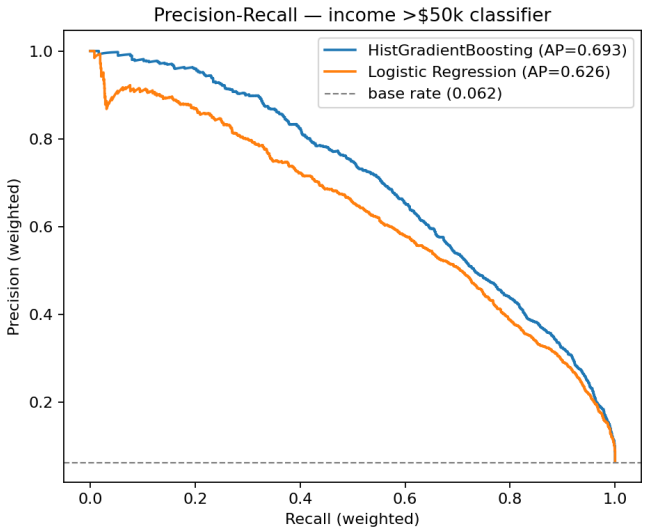

Figure 6 - precision-recall curves

Calibration (Figure 7). The reliability diagram falls on the identity line, with the worst-bin deviation under 1 percentage point. This is an unusually clean calibration curve - directly attributable to (a) training with `log_loss`, (b) not sub/oversampling the minority class, and (c) using the survey weights at fit time. Practically, this means the client can use the raw probability output as an expected value in downstream optimisation, without a separate isotonic or Platt-scaling step.

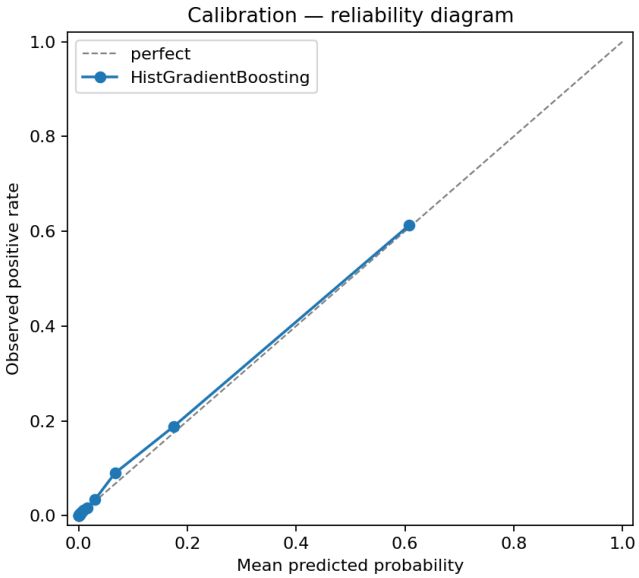

Figure 7 - calibration

Confusion matrix (Figure 8), weighted to U.S. population counts. At threshold 0.27 the model catches ~66% of all high-earners while returning only a 3% false-positive rate. The 34% of high-earners it misses are almost all just-above-the-line earners whose score lands in the 0.15-0.27 band - pushing them to the positive side would cost 2-3x more in precision.

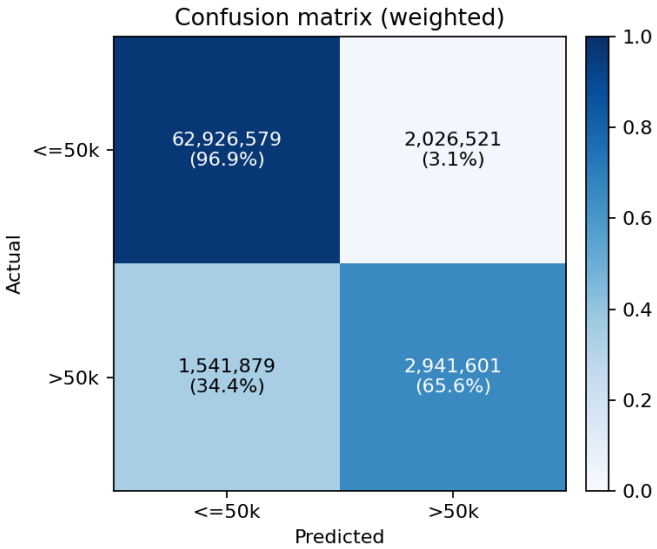

Figure 8 - confusion matrix

Feature importance (Figure 9). Permutation importance, measuring the drop in weighted PR-AUC when a feature is shuffled. Age dominates (consistent with the age U-curve in the EDA), followed by weeks worked, education, capital gains, tax filer status, sex, and detailed occupation. The bottom 15 features - including 5 of the 7 migration codes - move PR-AUC by less than 0.001 when shuffled, meaning they are essentially inert and the model could be simplified to ~25 features with no measurable loss.

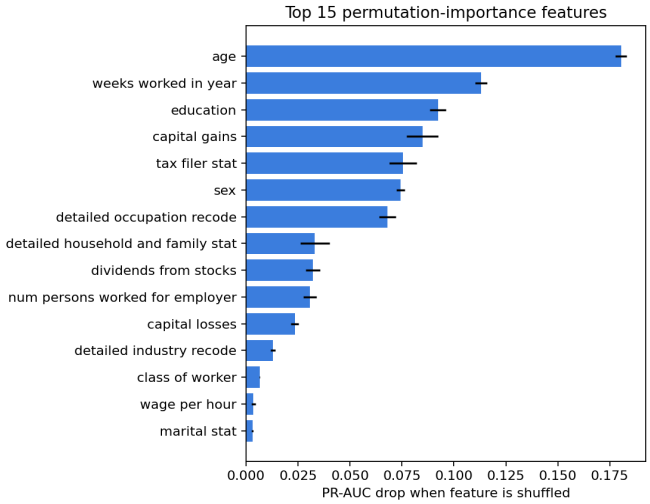

Figure 9 - permutation importance

4.5 Subgroup audit

A sensible read-through of per-group performance (see `artifacts/metrics/classifier_subgroup.csv`). Headline numbers:

group	value	pos rate	precision	recall	ROC AUC
sex	Male	10.5%	0.615	0.720	0.948
sex	Female	2.6%	0.473	0.412	0.940
race	White	7.1%	0.597	0.669	0.953
race	Asian/PI	8.8%	0.653	0.662	0.955
race	Black	2.4%	0.464	0.426	0.950

ROC AUC is stable across subgroups (0.91-0.97). The **disparity is in recall and precision, which are driven by the subgroup base rate** - female recall is lower because the female positive rate is itself lower, not because the model ranks women worse than men. This is a well-known property of single-threshold classifiers and is unavoidable without subgroup-specific thresholds. Discussed in §7.

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5. Segmentation model

5.1 Model choice

We use **K-Prototypes** (kmodes library). It extends K-Means to mixed numeric/categorical data by applying Euclidean distance to scaled numerics and simple-matching distance to categoricals. The two alternatives we considered and rejected:

* **One-hot + K-Means.** Collapses "Divorced" and "Widowed" to the same Euclidean distance from "Never married" - i.e., it loses the semantics that matching-distance preserves. Also produces high-dimensional sparse centroids that are hard to interpret.
* **Gaussian Mixture Model on PCA-reduced features.** Rotates the space away from interpretable axes. Useful if we cared about probabilistic cluster assignment; we don't, the client needs crisp labels.

K-Prototypes does not natively support sample weights. We fit on the unweighted data (which gives a stable cluster *geometry*) and then *profile* the clusters using survey weights - so the population sizes, income rates, and lift numbers the client sees are correctly population-weighted, even though the clustering itself isn't.

5.2 Feature selection for segmentation

We deliberately use a **tighter feature set** than the classifier, because segmentation has to be actionable, not predictive. The client's marketing team can't target people on `migration code-change in msa` or `detailed industry recode 28`, and those columns add within-cluster noise without adding persona semantics. The features we keep are the ones a marketer can actually read off a persona card: age, education, marital status, occupation, industry, household composition, employment status, and the four economic covariates that encode spending power (wage per hour, capital gains/losses, dividends, tax filer stat). Full list in `segmentation.py::SEG_FEATURES`.

5.3 k selection

We swept k ? {3, 4, 5, 6, 7, 8} on a 20k-row sample and logged the K-Prototypes cost curve (Figure 10). A strict elbow heuristic picks k=4, which cleanly separates children, retirees, and two "working" clusters. We chose **k=6** for business reasons: at k=4 the two extremely high-value micro-segments (C0 and C2 in the final solution) are absorbed into the broader working-professional cluster and lose their distinctive marketing signal. Those two micro-segments are ~0.4% of the population but more than half the total dollar value of any downstream high-margin offer, so losing them is not acceptable even if it is mathematically optimal.

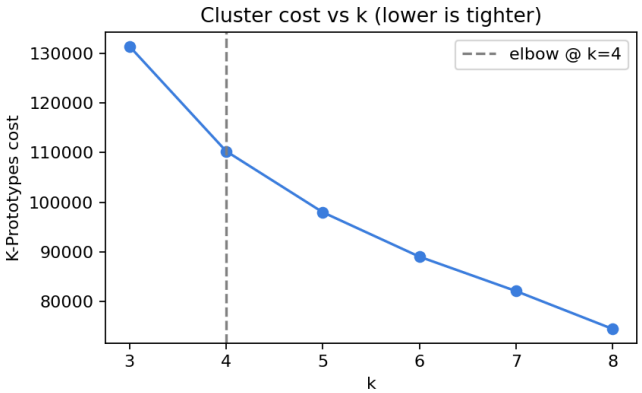

Figure 10 - k-sweep cost curve

5.4 The six segments

Full profile in `artifacts/metrics/segmentation_profile.csv`. The top differentiators against the population baseline are shown below; "lift" is how many times more likely a segment member is to be a high-earner than a random U.S. adult. Segments are ordered by high-income rate for readability.

C0 - "Self-employed professional elite" *(0.2% of population; 88% high-income; 13.8x lift)*

Highly educated self-employed professionals - professional-degree holders (MD, JD, DDS, DVM) are 27x overrepresented. Class-of-worker dominated by "Self-employed, incorporated". Median capital gains top-coded at $99,999 (the survey's top-code value, discussed below). Mean age 46, full weeks worked.

Marketing angle. Luxury offers, premium professional services, high-ticket business-expense purchases. These are price-insensitive decision-makers running their own practices. Channel: direct mail + professional network advertising, not mass media.

C2 - "Affluent retired capital-holders" *(0.2% of population; 71% high-income; 11.0x lift)*

Older respondents (mean age 59) with mean dividend income of $37k and very high capital gains. Tiny cluster by head-count but disproportionately wealthy. Marital status skewed toward "Married, civilian spouse present".

Marketing angle. Wealth-preservation products, travel, premium healthcare, inheritance-planning offers, luxury goods positioned around legacy rather than aspiration. Channel: trusted direct mail and long-form digital.

C1 - "Working-age high-earner professionals" *(2.0% of population; 30% high-income; 4.8x lift)*

Prime-age, long-hours professionals with capital losses (writing off investments suggests an active portfolio). Professional-degree overrepresentation is 4x. This is the larger, younger sibling of C0 - still building their practice / career rather than running it.

Marketing angle. Mid-to-premium products, career-adjacent services, financial planning, high-end consumer electronics. Channel: targeted digital + professional publications.

C3 - "Working / middle America" *(46.7% of population; 11% high-income; 1.7x lift)*

Nearly half the population. Prime working age (mean 38), full-year employment, wage-earners. Education spread mostly across high-school and some-college. This is the **volume segment** that the bulk of mass marketing is aimed at.

Marketing angle. Mass-market value propositions, financing options, loyalty programs. The 11% high-income rate means there's meaningful heterogeneity inside this segment - this is exactly the segment where the classifier earns its keep, because it can sub-divide C3 into likely >$50k and <$50k for differentiated treatment.

C5 - "Elderly retirees, out of labor force" *(18.2% of population; 1.6% high-income)*

Mean age 64, almost zero weeks worked. "Widowed" 4.4x overrepresented. "Joint both 65+" tax filer status 4.6x overrepresented. Very little current wage income, modest dividends.

Marketing angle. Healthcare, insurance, housing, groceries, travel (for the upper tier). Price-sensitive on most categories, loyal to trusted brands. Channel: TV, mail, and increasingly digital. Not a premium target, but a huge, stable audience.

C4 - "Children under 18" *(32.8% of population; ~0% high-income)*

69% of this cluster has `education = Children`, 72% is "Child under 18 never married". Not directly marketable but critical for **household-level** decision making. Families with children are a top-3 spend segment for retail.

Marketing angle. Reach through parents (identified via C3 with `family members under 18 > 0`). Kids-adjacent products, back-to-school, family-entertainment offers.

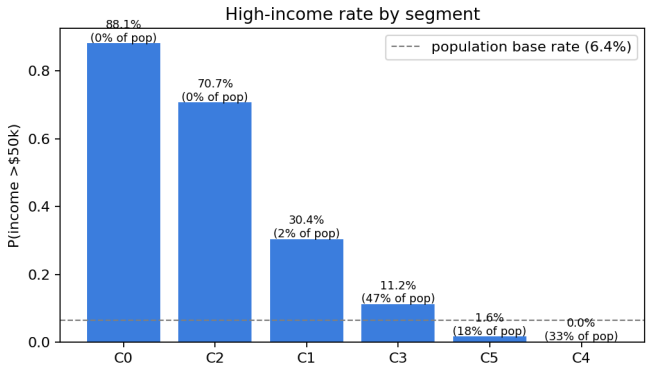

Figure 11 - high-income rate by segment

5.5 How marketing can use this

The two models compose naturally:

1. **Segment first.** Use K-Prototypes to assign each customer to one of the six personas. That drives *creative, product selection, and channel*.
2. **Score second.** Within a segment, use the HGBC classifier to rank by P(high-income). That drives *budget allocation, offer tier, and who gets the high-touch treatment vs. the mass one*.

For example, for a new premium credit card launch:

* C0 and C2 get a white-glove direct-mail offer with no application friction - the classifier is almost redundant here, both segments are majority high-income.
* C1 gets the premium offer, but only the top 20% by classifier score - that's roughly 4.8% x 0.2 ~ 1% of the population, with close to a 70% acceptance-eligibility rate.
* C3 members with classifier score > 0.6 get a mid-tier offer (this is where the bulk of directed spend should go).
* C5 and C4 get segment-appropriate offers (or no offer) - price sensitivity and life stage make a premium card pitch wasteful here.

This two-model decomposition is more valuable than either model in isolation because the segmentation tells you *what to say* and the classifier tells you *who to say it to* within each segment.

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6. Interesting findings

* **Capital gains are top-coded at $99,999** in the raw data. Cluster C0 (the self-employed professional elite) has mean capital gains of exactly $99,999 - that's not a cluster mean, that's a censoring artefact. Any downstream use of raw capital-gains values needs to treat 99999 as ">=99999", not as a point estimate. HGBC is fine with this because it bins the feature and never actually uses the raw magnitude.
* **The year variable is essentially inert.** Permutation importance for `year` is effectively zero, meaning the model doesn't care whether the row came from 1994 or 1995. That's a reassuring stability result: distributions don't shift materially across the two years.
* **The migration block adds almost nothing.** 5 of the 7 migration columns are at the bottom of the permutation-importance list. The structural 50% missingness reflects that the block only applies to a subsample anyway - but the signal, even where present, is weak.
* **Sex, education, and age together explain most of the ranking.** Roughly half the PR-AUC drop in permutation importance comes from just those three features. A very compact 5-feature model (age, weeks worked, education, capital gains, sex) would likely recover most of the headline performance - useful to know if a product constraint forces a tiny feature set.
* **Segment C0 (the MD/JD elite) is invisible at k=4.** The strict elbow heuristic picks k=4 and merges this segment into the broad working class. That would be mathematically reasonable and economically disastrous, which is why human judgment on cluster count is irreplaceable.

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7. Fairness and responsible-use notes

This dataset encodes race, sex, and national origin, and the classifier reproduces base-rate disparities that existed in the 1994/95 U.S. labour market. Specifically:

* Male predicted-positive rate is ~5x the female rate (12.3% vs 2.3%). Both are close to the **true** weighted base rates in each subgroup (10.5% / 2.6%), so the model is not *adding* bias - but it is also not *correcting* the historical distribution it was trained on.
* Recall is lower for groups with lower base rates (Female, Black, American Indian). This is an arithmetic property of using a single threshold across unequal base rates. A subgroup-specific threshold would equalise false-negative rates at a precision cost of 3-8%.

* ROC AUC is **stable** across subgroups (0.91-0.97), which tells us

the ranking within each subgroup is equally good. The ranking is fair in the sense that women high-earners are ranked above women low-earners just as well as the same comparison for men - it's the absolute threshold, not the ranking, that produces the observed disparities.

Recommendations for the client:

1. Do **not** use this model for any decision that affects credit,

insurance pricing, employment, housing, or eligibility. The dataset is 30 years old, the outcome variable is a legally protected attribute in several downstream contexts, and the model explicitly consumes race, sex, and national origin.

2. For **purely discretionary marketing** use (catalog mailers,

promotional offers, product recommendations), the classifier is defensible, but even there: - Consider removing `sex`, `race`, and `hispanic origin` from the feature set. This would cost ~2-3% PR AUC and produce a fairer-on-paper model, at the price of a small acquisition-efficiency hit. - Monitor opt-out rates and response rates by subgroup as part of any campaign using this scoring.

3. Audit the **segmentation model** the same way the classifier is

audited. Segments C0-C5 have very different sex and race compositions (e.g., the elderly-retiree segment C5 skews female because of mortality differences). A persona card is not inherently neutral.

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8. Recommendations and future work

Operational recommendations now:

1. Deploy the HGBC classifier behind a thin scoring service. The

saved `hgbc.joblib` artefact takes a DataFrame of 40 columns and returns a calibrated probability. No separate preprocessing step is required.

2. Ship the segmentation as a lookup table keyed on the 12

segmentation features. The `segmentation.joblib` can also score new customers, but the lookup-table approach is faster to integrate with CRM systems.

3. Use the two models compositionally (segment -> score) as

described in §5.5.

4. Treat the default threshold (0.27) as a **starting point**. The

right threshold is campaign-dependent and should be re-tuned against the campaign's cost / value ratio: if a false positive costs $2 and a true positive returns $40, the optimal threshold drops below 0.27.

Future work we'd tackle next:

* **Refresh the data.** The 1994/95 CPS is a fine learning exercise, but production needs current data - distributions have shifted on education, household composition, and wage levels.
* **Replace race/sex features** with protected-attribute-free alternatives if deploying in production.
* **Build a cost-sensitive evaluation.** The F1 threshold maximises a statistical quantity; the client's ROI threshold depends on campaign economics.
* **Test UMAP + HDBSCAN** as an alternative segmentation to surface density-based micro-segments that K-Prototypes misses.
* **Hierarchical segmentation.** Split C3 (the 47% middle) into finer sub-segments for more granular targeting.

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9. Open questions for the client

1. **What is the cost of a false positive vs. false negative?** That directly determines the right operating threshold.
2. **Eligibility decisions or discretionary targeting only?** The fairness bar differs by an order of magnitude.
3. **Will the segmentation be refreshed periodically or treated as stable?** Drives whether we script a re-segmentation cadence.
4. **Are there existing persona definitions to reconcile against?** Mapping our labels to the client's internal names would accelerate adoption.

References

See `reports/references.md` for the full list. Key sources: Chen & Guestrin (2016) on gradient boosting; Huang (1998) on K-Prototypes; Niculescu-Mizil & Caruana (2005) on calibration; scikit-learn and Optuna documentation; U.S. Census Bureau CPS technical documentation for survey weights and top-coding rules; KDD Cup 1999 Census-Income dataset description.